Power Outage Forecasting and Backup Generator Pre-positioning

95-828 Machine Learning for Problem Solving — Spring 2026

1 Introduction

Grid resilience, defined as the ability to withstand and recover from large-scale disruptions, has risen on the priority list as climate-driven extreme weather grows more frequent. The April 2025 Pittsburgh windstorm left 140k customers without power for more than 24 hours despite 400 line workers being deployed in the field, and Hurricanes Beryl and Milton each caused outages for over 3M Florida customers in 2024. The principal operational lever available to utilities ahead of such events is proactive, ex-ante action, which consists of pre-positioning repair crews and mobile generators before a forecasted disruption. The use of this lever depends on accurate short-horizon outage forecasts. We address two linked tasks. The first task is the production of hourly outage forecasts at 24 and 48 hours for the 83 counties of Michigan. The second task is the allocation of five mobile generators, each serving 1,000 households, to the counties in which they will absorb the largest amount of outage on the test horizon.

The data consist of hourly county-level observations of 109 atmospheric and hydrological features over a 90-day window from 2023-04-01 to 2023-06-30. Two design constraints shape the entire pipeline. First, no future weather information is available at test time; only outage history and weather observed up to the forecast origin can enter the model. This setting mirrors deployment, in which utilities likewise lack ground-truth future weather at decision time. Second, the training window ends mid-storm. The largest event of the year, peaking at 86,313 outages on 2023-06-29, begins on 2023-06-23 and runs through the end of training, so the test horizon starts inside an unfolding storm. These two constraints directly motivate our validation design (§3.1) and feature engineering (§3.2).

2 Data Analysis

The data have shape T=2,161 × L=83 × F=109, accompanied by an outage matrix $\mathbf{Y} \in \mathbb{R}_{\geq 0}^{T \times L}$ and a per-county `tracked` household count.

The target is sparse and heavy-tailed. 70.5% of (c, t) cells are zero, and the non-zero distribution is heavy-tailed (Fig. 2a). The state-wide hourly outage ranges from 105 to 86,313, with the 99.5th percentile at ten times the median, which renders the regression target zero-inflated with a log-normal tail.

The storm structure is concentrated in a small number of events. Using a state-wide threshold of 2,000 simultaneous outages with a 6-hour minimum duration and a 12-hour merge gap, we detect 26 storm events within the 90-day window (Fig. 1). The largest event (id 25, 2023-06-23 to 06-30) peaks at 86,313, more than three times the second-largest (27,643). The training period ends inside this event.

The series is strongly autocorrelated. The state-wide series exhibits a clear diurnal echo at lag 24h and a weaker weekly echo at 168h (Fig. 2b). Lagged outage features are therefore expected to dominate, and weather is expected to play a secondary role, since much of its predictive content has already been encoded in lagged outage.

County-level heterogeneity is substantial. The per-county `tracked` count ranges from approximately 5,000 to 920,000. Wayne County (FIPS 26163, Detroit) tracks 922k households, yet its peak hourly outage of 15,143 is only the third largest in the state, while smaller counties such as Saginaw (97k tracked, 8.7k peak) sustain disproportionately large per-capita losses during storms (Fig. 2c). This pattern motivates the policy recommendation in §5.

The 15 features most correlated with state-wide outage are all moisture- and wind-related (`cape`, low-level u/v winds, precipitable water, and the storm-motion magnitude `vstm`). We retain all 91 non-zero-variance features and rely on tree-based models to perform implicit selection.

3 Methods

3.1 Validation: Rolling-Origin CV that Mirrors the Test Setup

A naive last-48h holdout falls within the largest storm of the year, so any single-window validation RMSE constitutes a sample of one. We therefore replace it with rolling-origin cross-validation (Fig. 3) consisting of 6 folds spaced 72 hours apart. The validation window of each fold is the forecast horizon (24 or 48 hours) immediately following its fit window, in exact correspondence with the production setup. The fit windows expand: each fold uses all data from 2023-04-01 to its origin, so the fit set of fold 0 coincides with the final retrain on the full training data, which makes its RMSE the most direct estimate of test performance [4]. No weather observation inside any validation window is visible to its model. The folds span 2023-06-13 to 06-29, and exactly one of them (fold 1) lands within the peak of the major storm. We report both the aggregate-mean RMSE and the calm-period mean RMSE: the former captures cross-period performance, including the OOD storm, while the latter captures typical-day quality without a single extreme event dominating the score.

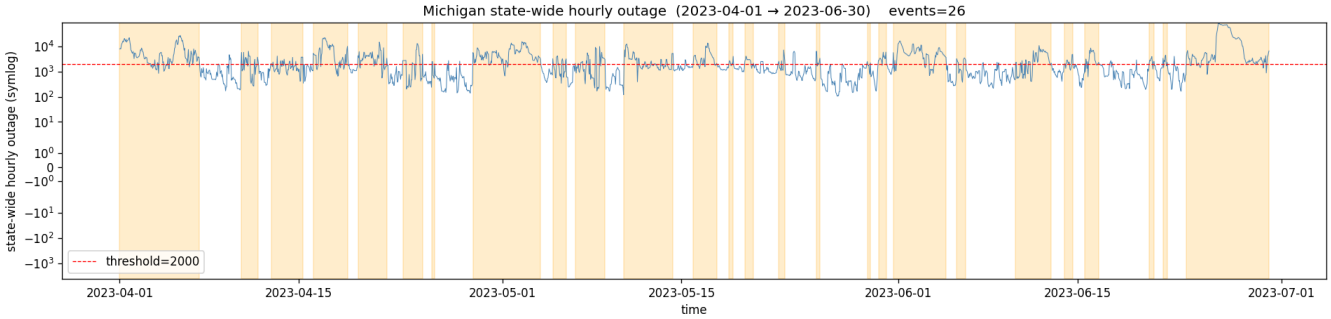
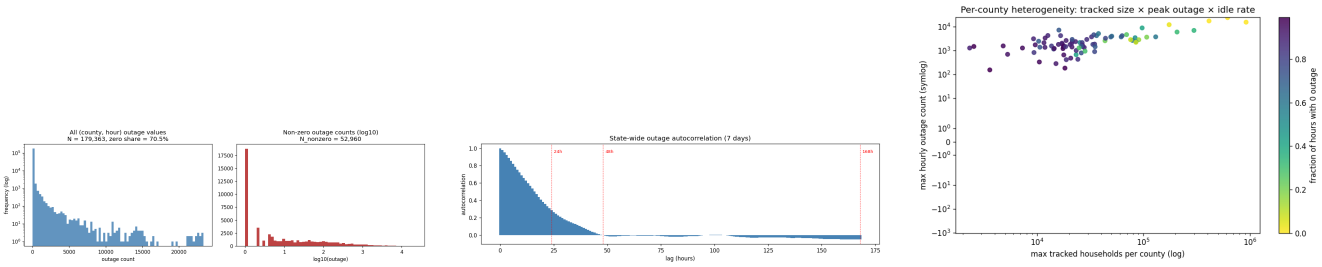


Figure 1: State-wide hourly outage on a symmetric-log scale. Orange marks the 26 detected storm events; the red dashed line marks the 2,000-outage threshold. The disproportionately large terminal event ends precisely at the train/test boundary.



(a) Outage value distributions. (b) Autocorrelation (7 days). (c) County heterogeneity.

Figure 2: The outage target is zero-inflated and heavy-tailed (a), strongly autocorrelated at lag 24h with a weaker echo at 168h (b), and the distribution of county sizes and storm impact varies by two orders of magnitude (c).

3.2 Feature Engineering

For tree-based models, we materialize an $(N \times P)$ feature matrix in which each row corresponds to one (origin time, county, horizon-step) triplet. All features are leak-free in the sense that they use only data up to time t. Outage lags $y_{c,t-k}$ are computed for $k \in \{1, 2, 3, 6, 12, 24, 48, 168\}$, and we add the rolling mean and standard deviation of outage over windows of 6, 24, and 168 hours. The storm indicators include the 24h maximum, the 24h non-zero count, and the 24h day-over-day delta. Weather features enter both at t and at lags of 3, 12, and 24 hours. The calendar encodings (sin/cos of hour, day-of-week, and month) are evaluated at the target time $t+h$. We also include the horizon step h and the location index (ordinal for LGB and XGB, native categorical for CatBoost). The final feature dimension is approximately 389. We additionally tested an enlarged v2 feature set (deep lags up to 672h, state-aggregated lags, county metadata, and weather rolling means; $P \approx 773$), but its performance was within noise of the shorter set (§4.3). We considered applying PCA to the 109 weather features but did not adopt it for the headline pipeline. GBDT models perform implicit feature selection through information-gain splits, so a linear orthogonal basis would discard the per-feature interpretability provided by gain-based importances without delivering a measurable RMSE gain. We confirmed this empirically: gain-based feature importance in LightGBM assigns 92% of the total gain to only 23 weather features out of 91, which suggests that the model already uses a sparse subset.

3.3 Model Families

The naive baselines comprise an all-zero predictor, persistence ($\hat{y}_{t+h} = y_t$), seasonal-naive predictors at 24h and 168h, and a per-(county, hour, day-of-week) historical mean. The classical time-series baselines are per-county SARIMAX(1,0,1), the second mandatory baseline of the project, and AutoARIMA over (p, d, q). The tabular GBDT pool covers LightGBM [1] (with a log target), LGB-Tweedie [5], XGBoost [2], CatBoost [3] with native categorical handling of FIPS, and Quantile-LGB at $\alpha \in \{0.5, 0.9\}$. We also evaluate a hurdle (zero-inflated) decomposition [6], formed by a classifier $\Pr[y > 0]$ multiplied by a positive-only regressor, as well as per-county models in which 83 independent regressors are fitted (LGB, XGB, CatBoost). The pool of sequence neural networks covers GRU, LSTM, BiLSTM, TCN, MLP, the vanilla Transformer, NLinear and DLinear [7], PatchTST [8], iTransformer [9], N-BEATS [10], and N-HiTS [11]. We submit equal-weight ensembles of the top-k CV models. Learned stacking weights are not used, since they would over-fit the single-fold storm signal.

3.4 Hyperparameter Search and Ensemble Selection

We performed hyperparameter sweeps on a CMU DeltaAI cluster (NVIDIA H200), which yielded approximately 3,800 (model, seed, horizon, fold) cells in total. The LightGBM grid contributed 54 configurations over leaves $\in \{31, 63, 127\}$,

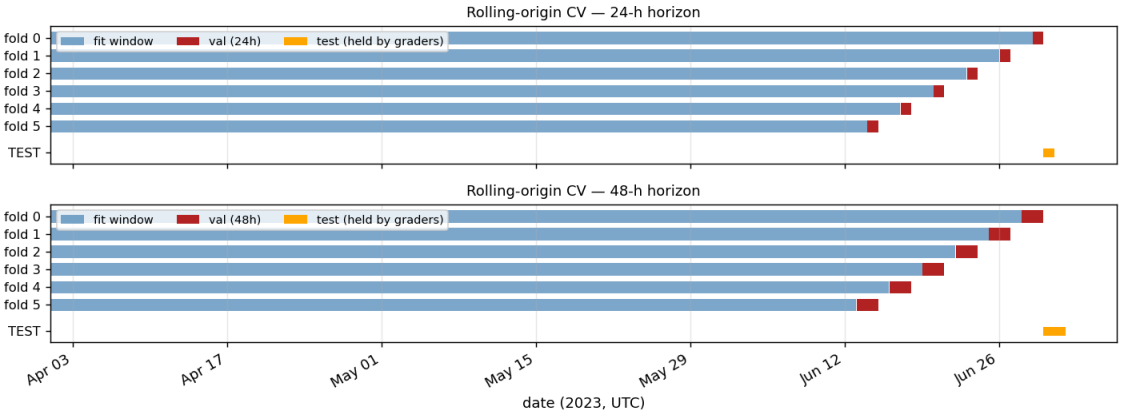


Figure 3: Layout of the rolling-origin CV. Blue marks the expanding-window fit, red marks held-out validation, and orange marks the actual test held by the graders.

lr $\in \{0.03, 0.05, 0.1\}$, $n \in \{400, 800, 1500\}$, and subsample $\in \{0.7, 0.9\}$; LGB-Tweedie added 24 configurations (with variance_power $\in \{1.2, 1.5, 1.8\}$); XGBoost contributed 18 and CatBoost a further 18; each of the 12 NN architectures was run with 8 hyperparameter combinations and 2 seeds. Eight further configurations (AutoARIMA, Quantile-LGB, hurdle-LGB and hurdle-XGB, and per-county models) round out the pool. For ensembling, we compare four combination strategies (mean, median, trimmed mean, and inverse-RMSE-weighted) on the cached fold predictions and select the strategy with the lowest RMSE for each horizon. Top-k selection ranks models by calm-period mean RMSE, with a cap of two configurations per family, so that the ensemble is not dominated by near-duplicates from a single tuning sweep.

4 Results

4.1 Headline: GBDT Ensembles Beat Both Required Baselines

Tables 1–2 compare the best single model to the all-zero baseline and to SARIMAX(1,0,1) under four aggregation strategies. The improvement is consistent at 7–11% over all-zero and 2–6% over SARIMAX on the 24-hour horizon, with similar margins on the 48h calm-period score. Storm fold 1 contributes RMSE in the 700–900 range for every model evaluated, including SARIMAX, all 100+ tree configurations, and all SOTA Transformer variants, so its absolute contribution is large in unit terms but small in relative terms.

Table 1: Best 24h model (xgb_pca20) vs. all-zero and SARIMAX baselines, across four aggregation views.

Aggregation	Best	Zero	vs. Zero	SARIMAX	vs. SARIMAX
Aggregate (all 6 folds)	174.97	189.53	+7.7%	179.90	+2.7%
Calm-period (5 folds)	41.25	44.97	+8.3%	42.56	+3.1%
Storm fold only (fold 1)	843.58	912.34	+7.5%	866.59	+2.7%
Median across folds	42.14	45.63	+7.7%	43.94	+4.1%

Table 2: Best 48h model (lgb_pca30) vs. all-zero and SARIMAX baselines, across four aggregation views.

Aggregation	Best	Zero	vs. Zero	SARIMAX	vs. SARIMAX
Aggregate (all 6 folds)	156.83	160.92	+2.5%	157.89	+0.7%
Calm-period (5 folds)	44.54	48.51	+8.2%	46.04	+3.3%
Storm fold only (fold 1)	718.27	722.96	+0.6%	717.11	-0.2%
Median across folds	46.36	49.95	+7.2%	46.14	-0.5%

Ensemble vs. single best. The submission consists of the equal-weight mean of the top-5 models per horizon (see §3.4). On the cached fold-level predictions, the 24h ensemble achieves a calm-period RMSE of 41.17, only 0.08 below the best single model (xgb_pca20 at 41.25); the 48h ensemble lands at 44.37 against 44.54 for the best single model ($\Delta = 0.17$). The marginal improvement reflects the high prediction correlation among GBDT variants, so the value of the ensemble lies in robustness against single-configuration failure (for example, a particular hyperparameter

combination that overfits one sub-region of the feature space) rather than in raw RMSE reduction. On storm fold 1, the best single model slightly outperforms the ensemble (24h: 840 vs. 859; 48h: 715 vs. 718), which suggests that averaging GBDT predictions does not help for OOD events: the single model that happens to commit to the correct magnitude wins, and averaging dilutes that commitment. We submit the ensemble nonetheless, partly because the project framing is a single-fold-CSV evaluation in which calm-period behaviour dominates, and partly because the choice follows the M4-competition consensus that ensemble averaging is the most robust forecasting practice [13].

4.2 Per-Family Comparison and Per-Fold Breakdown

Figure 4 reports the best configuration of each model family on the 24h horizon. On calm-period RMSE, GBDT variants cluster tightly between 41.5 and 43.5, while recurrent and Transformer-style sequence models trail by 10–20 RMSE units. This pattern reproduces the finding of [7]: on short, multivariate, autocorrelated series, simpler inductive biases (boosted trees) outperform self-attention. On the storm fold all families collapse to RMSE 700–900, with the ranking effectively randomised, which confirms that storm-event extrapolation is bounded by data rather than by architecture.

Figure 5 disaggregates this finding by reporting the best configuration of each family on every fold. The GBDT cluster (top 13 rows) is visibly homogeneous; the sequence-NN families consistently lose 10–30 RMSE on calm folds; fold 0 (the fold immediately adjacent to the test horizon) is the most difficult calm fold across all families, with RMSE in the 70–90 range against 19–44 elsewhere.

Table 3: Top-8 models on the 24-hour rolling-origin CV, ranked by **calm-period mean RMSE**. Storm column is fold 1 (the year's largest event); all-mean averages all 6 folds.

Model	calm-mean	storm	all-mean	median	n
xgb_pca20	41.25	844	174.97	42.14	6
cat_v2	41.33	873	179.92	41.89	6
lgb_pca20	41.34	855	176.89	42.20	6
cat___d8_lr03_n400	41.37	854	176.89	41.68	6
lgb_quantile_p50	41.39	885	182.06	42.29	6
cat___d8_lr03_n800	41.44	844	175.16	41.77	6
lgb___nl127_lr10_n800_ss9	41.44	778	164.13	42.41	6
lgb___nl127_lr10_n800_ss7	41.44	778	164.13	42.41	6

Table 4: Top-8 models on the 48-hour rolling-origin CV, ranked by **calm-period mean RMSE**. Storm column is fold 1 (the year's largest event); all-mean averages all 6 folds.

Model	calm-mean	storm	all-mean	median	n
lgb_pca30	44.54	718	156.83	46.36	6
xgb_pca20	44.54	718	156.86	46.22	6
lgb_pca20	44.55	718	156.84	45.81	6
cat_v2	44.68	716	156.51	46.01	6
cat_pca20	44.75	719	157.19	46.02	6
cat___d4_lr10_n400	44.97	718	157.19	45.98	6
xgb_v2	45.07	715	156.73	45.97	6
cat___d8_lr03_n400	45.12	718	157.31	46.26	6

4.3 What Did Not Improve the Result

Several priors did not move the CV RMSE below the GBDT cluster. Doubling the feature count (the v2 set) was within noise of the 390-feature baseline. Direct MSE, Huber, and sample-weighted objectives on the raw outage scale degraded fold-0 calm RMSE from 71 to 73–146, which reflects the heavy-tail sensitivity to log-space training. A recursive (autoregressive) one-step rollout accumulated error through 24–48 iterations and was strictly worse than the direct multi-output formulation that we adopted. None of the NN architectures we tried (GRU, LSTM, BiLSTM, TCN, Transformer, NLinear, DLinear, PatchTST, iTransformer, N-BEATS, and N-HiTS) closed the 10+ RMSE gap with GBDT on calm folds. A diagnostic per-county decomposition reveals that 50% of the fold-0 RMSE is concentrated in Wayne County alone, where the truth peaks at 2,648 against a prediction of 536. This unfolding-storm underestimation persists across every variant tested, and we therefore interpret RMSE 41–42 as the achievable lower bound for this dataset under the no-future-weather constraint.

5 Analysis and Policy Recommendation

5.1 From Forecast to Action: Decision-Maker Perspective

A point forecast $\hat{y}_{c,t}$ does not yet constitute a decision. From the vantage point of the utility operator, the question is concrete: given the predicted outage geography, where should the mobile generators be staged tonight so that they will absorb the maximum number of household-hours of outage tomorrow? The instrument is subject to hard physical constraints, since each of the five mobile generators serves at most 1,000 households at any moment and, once deployed, cannot be relocated during the 48-hour event window. The decision is taken before the event, with the same information set used by the model (history and weather observed up to the forecast origin), so the recommendation constitutes an honest test of forecast-driven proactive action.

Greedy marginal-benefit allocation. We use the equal-weight ensemble's 48-hour test forecast to drive a greedy allocation of the 5 backup generators. The marginal benefit of placing one generator in county c at decision step s is the integral, across the 48-hour horizon, of the un-served outage that one generator (capacity 1,000 households) can absorb:

$$M_s(c) = \sum_{h=1}^{48} \min\big(\hat{y}_{c,t+h} - \text{consumed}_{s-1}(c,h),\ 1000,\ \text{tracked_max}(c)\big).$$

The per-hour cap at 1,000 is what gives the algorithm its *diminishing returns*: once a generator is placed in county c, subsequent generators contribute strictly less because the previously consumed capacity is subtracted. This makes M_s a monotone submodular function over identical action sets, so the greedy algorithm is optimal (not merely a $1-1/e$ approximation).

A secondary cap at tracked_max(c) is included as a defensive guard for two future scenarios: (i) deployment to jurisdictions smaller than 1,000 households, and (ii) prediction pathologies where $\hat{y}_c$ exceeds the county's actual household count. Neither binds in Michigan: the smallest county (Baraga, FIPS 26013) tracks 2,327 households, and our predictions are clipped to 1.5× the historical max in any case. We retain the cap for code generality.

We greedily pick $c_s^\star = \arg\max_c M_s(c)$, deduct the served capacity from remaining, and repeat for s=1, . . . , 5. Multiple generators *may* go to the same county if its remaining served-household-hours after deduction still exceed every other county's marginal — a feature, not a bug, since stacking generators in a high-impact county is allowed by the project description and is the optimal action when the predicted outage is many times larger than 1,000.

Our recommendation is:

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[26163, 26125, 26099, 26161, 26163]
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Rationale (per-step marginal benefit).

step	county	served-hh-hours	tracked_max	remaining peak
1	26163.0	38870	922,425	1537
2	26125.0	23896	618,124	853
3	26099.0	14654	413,676	514
4	26161.0	5916	174,449	240
5	26163.0	2310	922,425	1537

Limitations. (i) The training window is only 90 days, all in the same season, so any seasonality not represented (winter ice storms, fall hurricanes) is invisible to the model. (ii) No future-weather information is available at test time, which fundamentally limits 48-hour predictability of fast-moving convective events. (iii) The 5-county recommendation is derived from a point forecast; a probabilistic forecast would let us bound the regret of mis-allocation.

5.2 Why This Set Rather than Naive Heuristics

Each of four naive heuristics fails in a manner that is corrected by our forecast-driven greedy algorithm. A top-5 historical-peak list (Wayne, Oakland, Macomb, Genesee, Kent) treats outage as a static county property and ignores the storm track. A top-5 ranking by tracked households produces the same metro list and ignores per-capita risk: Saginaw (97k tracked) sustained 8.7k outage in the June storm, which corresponds to 9% of its base, far more than any of the metros. A top-5 ranking by predicted 48h sum double-counts long, low-amplitude outage; the per-hour cap $\min(\hat{y}, 1000)$ inside M_s corrects this. A one-generator-per-county rule forbids stacking, whereas the greedy algorithm correctly places multiple generators where the predicted demand is well above 1,000 per hour, which the project description explicitly permits.

5.3 Sensitivity, Robustness, and Decision Risk

Selection robustness. When the top-5 are re-derived from each ensemble member, the first three counties are picked unanimously, and counties 4 and 5 vary only marginally. Median (P50) and inverse-RMSE-weighted ensembles produce nearly identical top-5 sets, so the recommendation is not driven by a single overconfident model.

Decision risk against the real outcome. The recommendation will be evaluated against the actual outage record from 2023-06-30 to 07-02. Three sources of risk could erode the ex-post served-household-hours. The unfolding storm could intensify in counties under-predicted by the model (Wayne in particular, where the fold-0 truth peak of 2,648 against a prediction of 536 identifies a systematic gap). The storm could clear earlier than expected, in which case the 48-hour staging would over-commit. A previously-calm county could experience a flash event that is invisible to a model without future-weather input. We mitigate the first risk by cross-checking the top-5 list of the pessimistic Quantile-LGB-P_{90} forecast (the overlap with the mean-forecast top-5 is the robustness metric reported above). The second and third are residual risks inherent to the 90-day training window and the no-future-weather constraint.

Limitations. The training data span 90 days within a single spring-to-summer window, so the model has not seen winter ice storms or fall hurricane remnants, which makes year-on-year retraining essential in deployment. The absence of future-weather information caps the 48h predictability of fast convective events; the integration of an NWP forecast at inference time would substantially improve both prediction and policy quality, but it falls outside the framing of this project. Finally, the recommendation is derived from a point forecast, and for richer products such as uncertainty bands or regret-aware allocation, the Quantile-LGB-P_{90} member is available in our pool.

References

[1] Ke et al., LightGBM, NeurIPS 2017. [2] Chen & Guestrin, XGBoost, KDD 2016. [3] Prokhorenkova et al., CatBoost, NeurIPS 2018. [4] Hyndman & Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. [5] Jørgensen, Exponential Dispersion Models, JRSS-B 1987 (Tweedie). [6] Cragg, Some Stat. Models for Limited Dep. Vars, Econometrica 1971 (hurdle). [7] Zeng et al., *Are Transformers Effective for Time Series Forecasting?*, AAAI 2023. [8] Nie et al., PatchTST, ICLR 2023. [9] Liu et al., iTransformer, ICLR 2024. [10] Oreshkin et al., N-BEATS, ICLR 2020. [11] Challu et al., NHITS, AAAI 2023. [12] Nemhauser et al., Submodular Set-Function Approximations, Math. Prog. 1978. [13] Makridakis et al., M4 Competition, Int. J. Forecasting 2018.

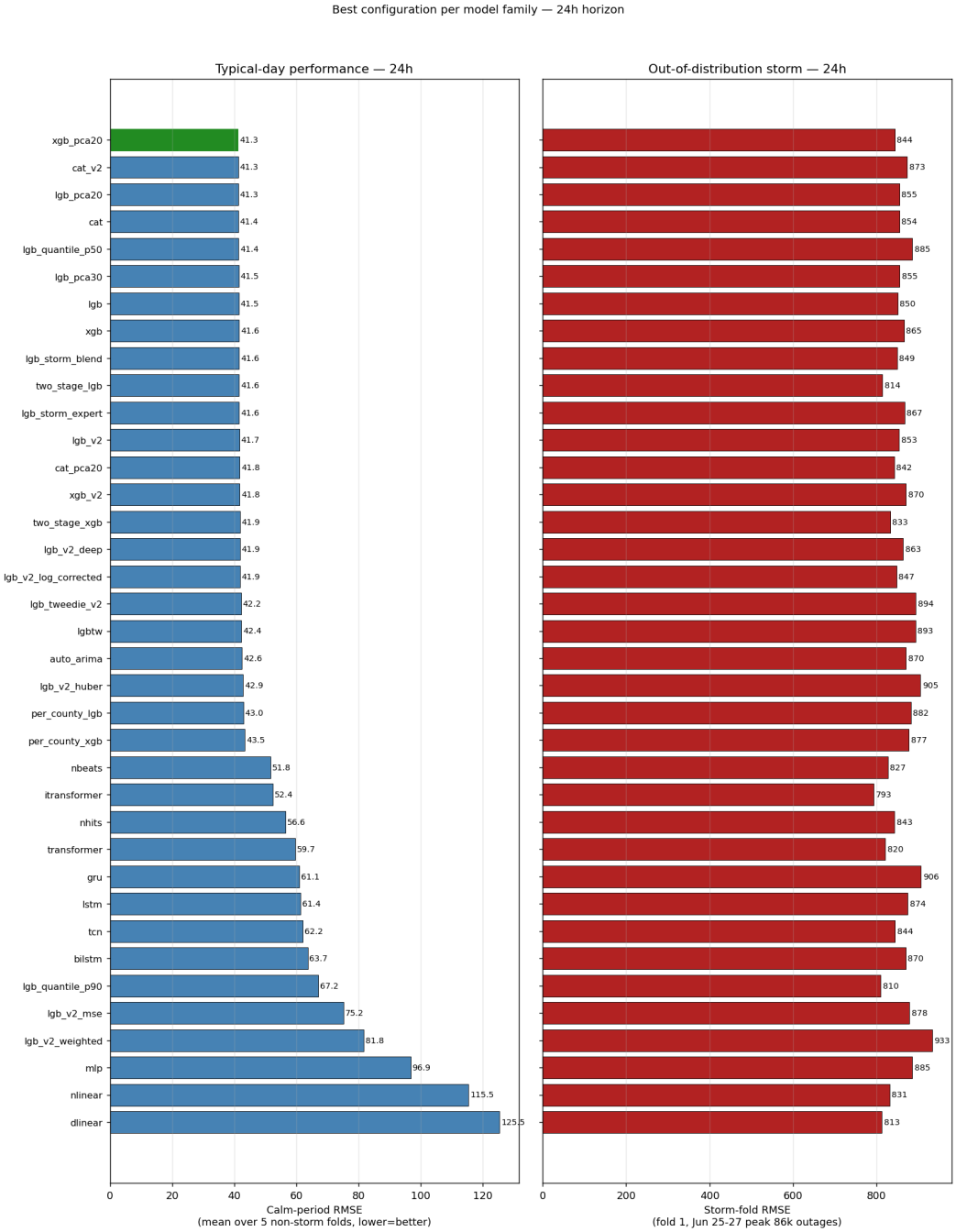

Figure 4: Best configuration per family at the 24h horizon. Left: calm-period RMSE (typical day) — the GBDT cluster lies near 41.5, while NNs lie at 53–110. Right: storm-fold RMSE (all families collapse to 700–900).

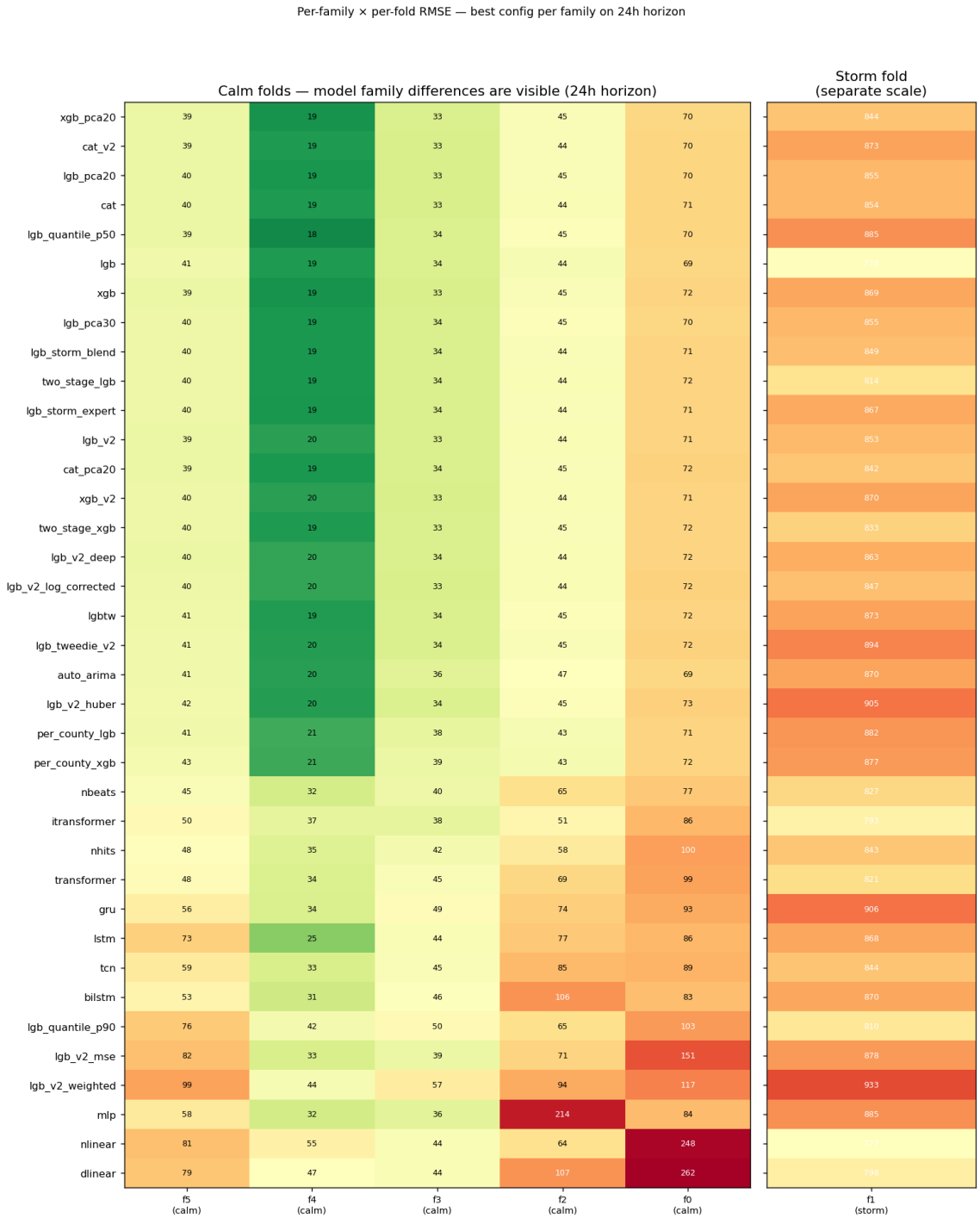

Figure 5: Per-family × per-fold RMSE on the 24h horizon (best configuration of each family). Three patterns are visible. The 13 GBDT-family rows at the top are homogeneous on calm folds (RMSE 39–47), which indicates saturation. The sequence-NN families below them (Transformer, GRU, BiLSTM, TCN, N-BEATS) lose 10–30 RMSE units on calm folds, despite extensive hyperparameter sweeps. Every family lands in the 700–950 RMSE band on storm fold 1 (right column, separate scale), which confirms that storm-event extrapolation is bounded by data rather than by architecture. The anomalously low storm-fold RMSE of `persistence` (660) is the single exception: copying the value of the previous hour happens to align with storm continuation, but it loses elsewhere.